

A 'MINORITY REPORT' STYLE GESTURE INTERFACE: Do We Really Need It?



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From text to graphics, from keyboards to mice to touchscreens, we have come a long way. Gestural interface comes next in this logical progression of human-machine interfaces (HMIs). The journey has already begun. We have advanced gesture sensors and software stacks to understand these gestures and use these as inputs for various applications. Products and projects like Microchip's GestIC, Nintendo Wii, Intel's RealSense, Leap Motion, Google's Soli, Microsoft's Kinect and Handpose, and Gest gloves do show some promise.

Many industry watchers also believe that the future of the Internet of Things (IoT) depends on the success of gesture based interaction because these devices will be very deeply intertwined in our lives, and we must be able to interact naturally with these if we are not to go crazy!

However, most of today's gesture inter-

faces have some constraints. Many work together with touch and not independently. Others might require you to sit in front of the computer within the acceptable range, wear a gesture-control device, ensure that the room is properly lit and probably even train the computer to your hand size and movements.

Recall scenes from the movie *Minority Report* and you begin to wonder whether the existing technologies are anywhere close to that kind of intuitive gesture control. Can you grab files in the air and throw these into folders? Would it be possible to sift through and organise tonnes of data within minutes? Would it be possible to use gestures to simulate a plan and implement it?

There is abundant research happening around the world to enable all this, and we will probably be interacting with computers using purely gestures sometime in the future. However, would that really be necessary for all of us?

Not all tech works for all devices

Some experts feel that gestures might not be needed for all kinds of computing. Take the example of touchscreens. While initially not very successful in larger computers, these have evolved into *de facto* tech for smartphones and tablets. Gesture based interfaces too might make it big only in such specialised applications. For example, handling files on a laptop or doing something on the phone are more of personalised tasks. You do not want to be gesturing to such personal devices, drawing everybody's attention to what you are doing.

However, in more collaborative environments, or when the user has disabilities that hamper the use of regular user interfaces (UIs), or under demanding environ-

Fig. 1: Tongue-n-Cheek, a non-contact tongue-gesture-recognition system (Courtesy: University of Maryland)

